

LAUNCESTON AP, Australia

WMO: 959660

Lat: 41.5477S

Lon: 147.2160E

Elev: 168

StdP: 99.32

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.2	-0.1	-3.7	2.8	5.4	-2.8	3.1	4.7	13.2	10.9	12.0	10.5	2.0	190	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	28.2	17.6	26.1	17.0	24.3	16.1	19.1	25.1	18.2	23.7	17.3	22.5	7.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.5	20.4	16.1	11.7	20.0	15.1	11.0	19.4	55.1	25.2	52.2	23.7	49.4	22.5	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.5	10.4	9.4	-3.6	32.2	0.8	2.7	-4.2	34.1	-4.7	35.7	-5.1	37.2	-5.8	39.2
			-3.8	20.9	0.8	1.0	-4.4	21.5	-4.9	22.1	-5.4	22.6	-6.0	23.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	17.2	17.1	15.2	11.9	9.4	7.4	6.9	7.6	9.1	11.0	13.2	15.2
	DBStd	4.44	3.02	2.85	2.99	2.59	2.49	2.37	2.11	2.14	2.14	2.45	2.74	2.78
	HDD10.0	378	0	0	1	11	42	83	99	79	41	17	4	1
	HDD18.3	2464	57	52	104	192	278	329	356	333	276	228	157	103
	CDD10.0	1013	224	199	163	69	23	5	2	5	15	47	99	162
	CDD18.3	57	23	18	8	0	0	0	0	0	0	0	1	6
	CDH23.3	629	261	162	77	2	0	0	0	0	0	3	28	97
	CDH26.7	144	73	31	16	0	0	0	0	0	0	0	3	21
Wind	WSAvg	4.6	5.0	4.6	4.4	4.1	4.1	3.9	4.2	4.7	4.8	4.9	4.9	5.1
Precipitation	PrecAvg	652	45	34	39	52	59	61	72	76	63	53	49	48
	PrecMax	939	154	164	100	183	129	169	188	223	121	133	100	128
	PrecMin	410	8	0	2	5	4	9	21	12	8	10	16	0
	PrecStd	138	31	33	26	36	35	30	37	41	28	28	21	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	30.1	28.8	22.6	18.0	15.0	14.0	15.7	18.6	22.9	26.4	29.3
		MCWB	19.3	18.5	18.0	15.3	13.2	11.8	10.6	11.2	12.8	14.6	16.5	17.4
	2%	DB	28.6	27.2	25.2	20.2	16.2	13.7	12.9	14.1	16.4	19.9	23.4	25.9
		MCWB	17.7	17.9	16.8	14.3	12.4	11.0	9.8	10.3	11.5	13.4	15.5	16.5
	5%	DB	26.1	25.2	23.1	18.7	15.0	12.8	11.9	12.9	15.1	18.0	21.0	23.6
		MCWB	16.9	16.9	15.7	13.4	11.8	10.5	9.3	9.6	10.8	12.6	14.4	15.3
	10%	DB	24.0	23.4	21.2	17.2	14.0	11.8	11.0	11.9	14.0	16.4	19.1	21.8
		MCWB	16.0	16.0	14.7	12.7	11.3	9.8	8.9	9.2	10.2	11.6	13.2	14.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	20.2	19.2	16.7	14.4	13.0	11.5	12.2	13.9	15.8	18.2	19.0
		MCDB	28.3	25.7	25.9	20.0	16.1	13.9	13.1	14.2	16.5	20.0	23.9	25.0
	2%	WB	19.1	19.1	17.8	15.4	13.5	11.7	10.6	11.2	12.6	14.4	16.5	17.7
		MCDB	25.4	24.5	22.9	18.7	15.2	13.0	11.9	13.1	15.4	18.5	21.5	23.1
	5%	WB	18.0	18.2	16.9	14.5	12.7	10.9	10.0	10.4	11.6	13.3	15.3	16.5
		MCDB	23.9	23.4	21.4	17.6	14.4	12.1	11.1	12.1	14.2	17.0	19.8	22.0
	10%	WB	16.9	17.1	15.8	13.5	11.8	10.2	9.3	9.6	10.6	12.2	14.2	15.3
		MCDB	22.5	22.0	19.8	16.4	13.6	11.5	10.6	11.3	13.1	15.8	18.4	20.6
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.0	11.8	10.5	9.2	8.3	8.3	9.0	9.7	10.8	11.5	12.4
		MCDBR	15.6	14.0	13.7	12.2	9.5	8.5	8.6	9.7	11.0	12.6	14.1	14.8
		MCWBR	6.7	6.2	6.4	7.1	6.4	6.2	6.2	6.5	7.0	7.4	7.6	7.1
	5% WB	MCDBR	13.8	12.6	12.3	10.6	7.9	7.4	7.2	8.1	10.0	11.5	12.4	13.3
MCWBR		6.9	6.5	6.3	6.8	5.8	5.8	5.7	6.0	7.0	7.2	7.4	7.3	
Clear-Sky Solar Irradiance	taub	0.338	0.331	0.323	0.312	0.302	0.289	0.289	0.305	0.325	0.331	0.333	0.333	
	taud	2.502	2.542	2.556	2.563	2.567	2.590	2.599	2.550	2.483	2.475	2.494	2.508	
	Ebn at Noon	989	971	934	875	815	796	824	870	916	960	988	1000	
	Edh at Noon	111	101	91	78	65	58	62	77	96	107	111	112	
All-Sky Solar Radiation	RadAvg	6.95	5.85	4.52	3.06	1.98	1.53	1.70	2.50	3.71	5.17	6.21	6.89	
	RadStd	0.48	0.24	0.20	0.16	0.12	0.13	0.08	0.20	0.25	0.27	0.46	0.45	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.30	N/A	+0.33	+0.90	N/A	N/A	N/A	-98	+93
(47)	Regional (34 neighbors)	+0.32	N/A	N/A	+0.68	+0.36	+0.43	N/A	-100	+108

Nomenclature: See separate page